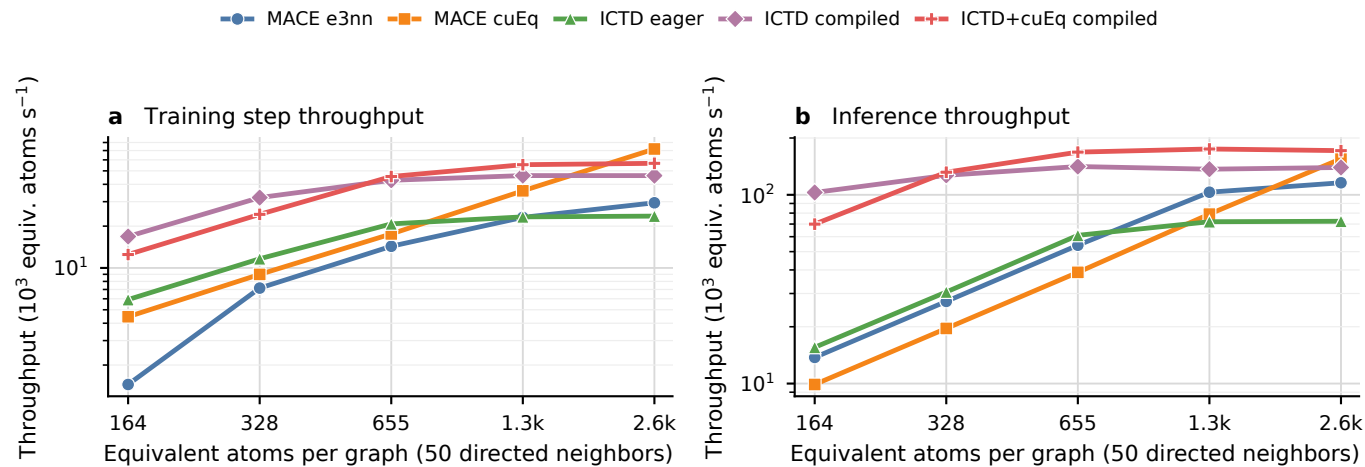




RTX 4090, FP32, 64 channels, avg. directed degree 16. Each point is the fastest successful run over $(\ell_{\text{hidden}}, \ell_{\text{max}}) \in \{(1, 2), (2, 3)\}$; compiled = make_fx for training and AOTI for inference. Atom counts/throughput are converted from directed edges assuming 50 directed neighbors per atom.